

Lecture 5: Conditionals



INFO 1100: Introduction to Programming

July 4th?

Survey Check-In

Survey Check-In

How useful are course materials?

Survey Check-In

How useful are course materials?

Readings: **Agree (4.2)**

GRQs: **Strong-ish Agree (4.5)**

Lectures: **Agree-ish (3.6)**

Labs: **Strong-ish Agree (4.6)**

Assignments: **Agree (4.4)**

Survey Check-In

How useful are course materials?

Nice!

Readings: **Agree (4.2)**

GRQs: **Strong-ish Agree (4.5)**

Lectures: **Agree-ish (3.6)**

Labs: **Strong-ish Agree (4.6)**

Assignments: **Agree (4.4)**

Survey Check-In

How useful are course materials?

What could
make lectures
more helpful?

Readings: **Agree (4.2)**

GRQs: **Strong-ish Agree (4.5)**

Lectures: **Agree-ish (3.6)**

Labs: **Strong-ish Agree (4.6)**

Assignments: **Agree (4.4)**

Survey Check-In

How well is the course paced?

Survey Check-In

How well is the course paced?

Very slightly too fast

3.6

Survey Check-In

Other comments:

Survey Check-In

Other comments:

People really like the
mixture of assignments
(labs, GRQs, etc)

Survey Check-In

Other comments:

People really like the
mixture of assignments
(labs, GRQs, etc)

... but there's a
lot of them!

Survey Check-In

Other comments:

People really like the
mixture of assignments
(labs, GRQs, etc)

... but there's a
lot of them!

Remember completion!
And talk to me!

Survey Check-In

Other comments:

Sometimes instructions
are unclear.

Survey Check-In

Other comments:

Sometimes instructions
are unclear.

What've been the snags?

Survey Check-In

Other comments:

More practice would
be helpful!

Survey Check-In

Other comments:

More practice would
be helpful!

Automate the Boring Stuff by Al Sweigart

futurecoder.io

CodingBat

...

Conditionals

Conditionals

Why do we need
conditionals?

Conditionals

so we can **make**
decisions!

Conditionals

```
print("Today you should wear:")  
#   > 60: no jacket required  
# 40-60: jacket  
# 20-40: winter coat  
#   < 20: everything you own...
```

Conditionals

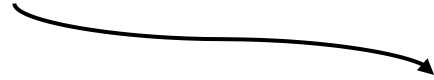
we need to be able to
control what the code
does based on **input**

If-Statements

```
if <boolean-expression>:  
    <something>  
  
    ...  
  
    <something>
```

If-Statements

if <boolean-expression> is True



```
if <boolean-expression>:
```

```
    <something>
```

```
    ...
```

```
    <something>
```


If-Statements

if <boolean-expression> is True

The diagram illustrates the mapping from a high-level 'if' statement to a low-level 'if-then' structure. A curved arrow points from the '<boolean-expression>' in the high-level statement to the '<boolean-expression>' in the low-level statement. Another set of three curved arrows points from the word 'then' to the three lines of code in the low-level statement: '<something>', '...', and '<something>'.

```
if <boolean-expression>:  
    <something>  
    ...  
    <something>
```

Boolean Expressions

what is a **Boolean expression**?

Boolean Expressions

what is a **Boolean expression**?

Take a stab?

Boolean Expressions

what is a **Boolean expression**?



an expression that is either
True or **False**

Boolean Expressions

often include **relational operators**

Boolean Expressions

often include **relational operators**

==	# is equal to
!=	# not equal to
>	# greater than
<	# less than
>=	# greater than or equal to
<=	# less than or equal to

Boolean Expressions

often include **relational operators**

Enter in the chat:

When might we
use these?

Boolean Expressions

often include **logical operators**

Boolean Expressions

often include **logical operators**

x **and** y

Boolean Expressions

often include **logical operators**

x **and** y



True only if x and y are True

Boolean Expressions

often include **logical operators**

$x \text{ or } y$

Boolean Expressions

often include **logical operators**

$x \text{ or } y$



True only if x or y are True

Boolean Expressions

often include **logical operators**

not x

Boolean Expressions

often include **logical operators**

not x



True only if x is False

Boolean Expressions

often include **logical operators**

Enter in the chat:

When might we
use these?

Boolean Expressions

examples of **Boolean expressions**

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Boolean variables

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Boolean variables

```
if is_rainy:
```

```
    print("Umbrella!")
```

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Boolean operations

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Boolean operations

```
if is_windy and not is_rainy:  
    print("Fly a kite!")
```

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Comparison operations

Boolean Expressions

examples of **Boolean expressions**

```
is_rainy = False
```

```
is_windy = True
```

```
temp = 12
```

Comparison operations

```
if temp < 30 and is_rainy:  
    print("Beware of ice!")
```

If-Statements

```
a = 0  
  
if a == 0:  
    a = a + 1  
  
if a == 0:  
    a = a + 2  
  
a = a + 1  
  
print(a)
```

What's printed?

Conditionals

What if we want
multiple conditions?

If-Else Statements

```
if <boolean-expression>:  
    <something>  
else:  
    <something>
```

If-Else Statements

```
if <boolean-expression>:  
    <something>  
else:  
    <something>
```

How do I work?

If-Else Statements

Enter in the chat:

What's one situation
where you might use
an **if / else** statement?

Conditionals

What if we want **even
more** conditions?

Conditionals

```
print("Today you should wear:")  
#   > 60: no jacket required  
# 40-60: jacket  
# 20-40: winter coat  
#   < 20: everything you own...
```

Elif Statements

```
if <boolean-expression>:  
    <something>  
elif <boolean-expression>:  
    <something>  
else:  
    <something>
```

Elif Statements

```
if <boolean-expression>:  
    <something>  
elif <boolean-expression>:  
    <something>  
else:  
    <something>
```

How do I work?

Elif Statements

```
if temp > 60:  
    print("No jacket")  
elif temp > 40:  
    print("Jacket")  
elif temp > 20:  
    print("Coat")  
else:  
    print("All the gear!")
```

Elif Statements

```
if <boolean-expression>:
```

```
    <something>
```

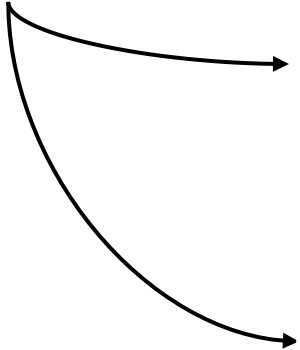
```
elif <boolean-expression>:
```

```
    <something>
```

```
else:
```

```
    <something>
```

optional



Elif Statements

**as many
as you
want**

```
if <boolean-expression>:
```

```
    <something>
```



```
elif <boolean-expression>:
```

```
    <something>
```

```
else:
```

```
    <something>
```

Elif Statements

**as many
as you
want**

```
if <boolean-expression>:
```

```
    <something>
```

 `elif <boolean-expression>:`

```
    <something>
```

```
else:
```

```
    <something>
```

**but must be after
if + before else**

Elif Statements

```
if <boolean-expression>:
```

```
    <something>
```

```
elif <boolean-expression>:
```

```
    <something>
```

```
else:
```

```
    <something>
```

**expressions are
checked in order**

Elif Statements

```
if <boolean-expression>:
```

```
    <something>
```

```
elif <boolean-expression>:
```

```
    <something>
```

```
else:
```

```
    <something>
```

**expressions are
checked in order**

once Python finds a True
Boolean expression, it
skips all others

Elif Statements

Enter in the chat:

What's one situation where you might use an **elif** statement?

Elif Statements

```
x = 0
```

```
if x == 0:
```

```
    print("x is 0!")
```

```
if x == 0:
```

```
    print("still 0!")
```

```
if x == 0:
```

```
    print("even still 0!")
```

what prints?

Elif Statements

```
x = 0
```

```
if x == 0:
```

```
    print("x is 0!")
```

```
elif x == 0:
```

```
    print("still 0!")
```

```
elif x == 0:
```

```
    print("even still 0!")
```

what prints?

Nested Conditionals

```
if <b-e>:  
    if <b-e>:  
        <statement>  
    elif <b-e>:  
        <statement>  
else:  
    <statement>
```

Nested Conditionals

**sometimes
make things
cleaner...**

```
if <b-e>:  
    if <b-e>:  
        <statement>  
    elif <b-e>:  
        <statement>  
else:  
    <statement>
```

Nested Conditionals

```
if <b-e>:  
    if <b-e>:  
        <statement>  
    elif <b-e>:  
        <statement>  
else:  
    <statement>
```

**sometimes
make things
cleaner...**

**sometimes
messier...**

Nested Conditionals

```
if raining and freezing:
    print('Waterproof coat.')
elif raining and not freezing:
    print('Umbrella.')
elif not raining and freezing:
    print('Warm coat!')
else:
    print('Sweater.')
```

```
if raining:
    if freezing:
        print('Waterproof
coat.')
    else:
        print('Umbrella.')
else:
    if freezing:
        print("Warm coat!")
    else:
        print('Sweater.')
```

Nested Conditionals

```
if raining and freezing:
    print('Waterproof coat.')
elif raining and not freezing:
    print('Umbrella.')
elif not raining and freezing:
    print('Warm coat!')
else:
    print('Sweater.')
```

```
if raining:
    if freezing:
        print('Waterproof
coat.')
    else:
        print('Umbrella.')
else:
    if freezing:
        print("Warm coat!")
    else:
        print('Sweater.')
```

Which do you prefer?

Variables in conditionals

Variables created inside an `if` statement continue to exist outside of the `if`

Variables in conditionals

```
a = 0
if a == 0:
    b = a + 1
print(b)
```


Variables in conditionals

```
a = 0
if a == 0:
    b = a + 1
print(b)
```

What prints?

Variables in conditionals

```
a = 3  
if a == 0:  
    b = a + 1  
print(b)
```

Variables in conditionals

```
a = 3
if a == 0:
    b = a + 1
print(b)
```

What prints?

Questions?

(... I'll wait for 3)

For tomorrow:

(07/01 @ 11:30am)

- Lab 5
- Readings
Think Python –
3.1-3.3, 3.8-3.9, 6.1-6.3
- GRQs